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COMPUTER

CHAPTER 14

Computer Network Models

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CHAPTER 14

Computer Network Models

Computer Awareness

IN THIS CHAPTER

- OSI model
- Seven OSI layers
- Network devices and layers
- TCP/IP model
- Network protocols
- OSI versus TCP/IP

COMPUTER NETWORK MODELS

A Network Model is a framework that describes how data is transmitted from one computer to another over a network.

The two most important network models are:

Computer Network Models

- OSI Model
- TCP/IP Model

OSI MODEL(OPEN SYSTEMS INTERCONNECTION)



OSI layer order, from user services to transmission.

The OSI Model is a reference model that explains how data travels from a software application on one computer to a software application on another computer through a network.

It divides the communication process into seven layers, with each layer performing a specific function.

Developed By ISO (International Organization for Standardization) in Year 1984

Why OSI Model was Developed?

- To make communication between different systems easier.

FEATURES OF OSI MODEL

- Consists of 7 Layers
- Each layer performs a specific function.
- Layers work independently.
- Simplifies network communication.

UPPER LAYERS AND LOWER LAYERS

Upper Layers

These layers deal mainly with software and user-related functions.

- Application Layer
- Presentation Layer
- Session Layer

Function

Provide services directly to users and applications.

Lower Layers

These layers deal with data transmission and hardware.

- Transport Layer
- Network Layer
- Data Link Layer
- Physical Layer

Function

Handle data movement across the network.

OSI MODEL LAYERS

7. Application Layer 6. Presentation Layer

5. Session Layer

4. Transport Layer

3. Network Layer

2. Data Link Layer

1. Physical Layer

DATA UNITS IN OSI MODEL

Layer	Data Unit
Application	Data
Presentation	Data
Session	Data
Transport	Segment
Network	Packet
Data Link	Frame
Physical	Bits

1. PHYSICAL LAYER

The Physical Layer is the lowest layer of the OSI Model.

It is responsible for transmitting individual bits from one device to another.

Main Functions

1. Data Transmission

Defines how bits are transmitted.

2. Transmission Mode

Determines whether communication is:

- Simplex
- Half-Duplex
- Full-Duplex

3. Topology

Defines the arrangement of devices in a network.

4. Signal Type

Determines the type of signal used for communication.

Devices

- Hub
- Repeater

Data Unit:Bits

Exam Point

Physical Layer deals with Bits and Transmission Media.

2. DATA LINK LAYER

The Data Link Layer transfers frames across a local link and may detect transmission errors; it does not guarantee error-free delivery.

It converts raw bits into frames.

Main Functions

1. Framing

Converts bits into Frames.

2. Error Detection

Detects transmission errors.

3. MAC Addressing

Provides hardware addressing.

4. Reliable Communication Ensures data is transferred correctly.

Protocols used in this layer:

- IEEE 802.3
- CSMA/CD(carrier sense multiple access /collision detection)
- CRC(cyclic redundancy check)
- HDLC (High-Level Data Link Control)

Devices

- Switch
- Bridge

Data Unit:Frame

Exam Point

Data Link Layer uses MAC Addresses.

FRAMING

Framing is the process of dividing data into units called Frames

3. NETWORK LAYER

The Network Layer is responsible for routing and logical addressing.

It determines the best path from source to destination.

Main Functions

1. Internetworking

Provides communication between different networks.

2. Addressing

Uses IP addresses to identify devices.

3. Routing Selects the best path for data transmission.

4. Packetizing

Converts data into packets.

Protocols used in this layer:

- IP (Internet Protocol)
- IPv6 (Internet protocol version six)
- ICMP(Internet control message protocol))
- NAT (Network Address Translation)

Device :Router

Data Unit :Packet

Exam Point :Router works at Network Layer (Layer 3).

Memory Trick

Network Layer = IP Address + Routing

4. TRANSPORT LAYER

The Transport Layer provides end-to-end transport. TCP offers reliable delivery; UDP does not guarantee delivery.

Protocols used in this layer:

1.TCP (Transmission Control Protocol)

TCP is a reliable, connection-oriented protocol.

Working

1.Data is divided into segments.

2.Segments travel through different routes. 3.Receiver reorders them correctly.

Example

- File Download
- Email Transmission

2. UDP (User Datagram Protocol)

UDP is a connectionless and unreliable protocol.

Features

- Faster communication

Example

- Live Video Streaming
- Online Gaming

- VoIP Calls

5. SESSION LAYER

The Session Layer establishes, manages, and terminates communication sessions between devices.

Protocols used in this layer:

- PAP (Password Authentication Protocol)
- RPC (Remote Procedure Call)

6. PRESENTATION LAYER

The Presentation Layer acts as a translator between different systems.

It is also called the Syntax Layer Main Functions

1. Translation

Converts data from sender format to receiver format.

2. Encryption

Protects data from unauthorized access.

3. Compression

Reduces data size for faster transmission.

Protocols used in this layer:

- SSL (Secure Sockets Layer)
- TLS (Transport Layer Security)
- MIME (Multipurpose Internet Mail Extensions)

Exam Point

Presentation Layer = Translation + Encryption + Compression

7. APPLICATION LAYER

The Application Layer is the topmost layer of the OSI Model.

It provides network services directly to users and applications.

Functions

1. File Transfer

Transfer files between computers.

2. Mail Services

Supports email services.

IMPORTANT PROTOCOLS OF APPLICATION LAYER

Protocol	Full Form	Purpose
HTTP	HyperText Transfer Protocol	Web Browsing
FTP	File Transfer Protocol	File Transfer
SMTP	Simple Mail Transfer Protocol	Sending Email

OSI LAYER DEVICES

Device	Layer
Repeater	Physical
Hub	Physical
Bridge	Data Link
Switch	Data Link
Router	Network
Gateway	Depends on function; a default IP gateway is normally a network-layer router.

Exam Point

Router = Layer 3 Device

Switch = Layer 2 Device

MOST IMPORTANT FACTS FOR EXAMS

1.OSI stands for Open Systems Interconnection.

2.OSI Model was developed by ISO in 1984.

3.OSI has 7 layers.

4.Physical Layer deals with Bits.

5.Data Link Layer deals with Frames.

6.Network Layer deals with Packets.

7.Transport Layer deals with Segments.

8.Router works at Layer 3.

9.Switch works at Layer 2.

10.TCP is reliable. 11.UDP is unreliable.

12.Presentation Layer handles Encryption and Compression.

13.Application Layer is closest to the user.

14.NAT converts private IP to public IP.

15.HTTP is used for web browsing.

TCP/IP MODEL (TRANSMISSION CONTROL PROTOCOL /

INTERNET PROTOCOL)

The TCP/IP Model is the standard networking model used on the Internet.It defines how data is transmitted between computers over a network.

The TCP/IP Model was developed before the OSI Model by the ARPANET (Advanced Research Projects Agency Network) project, which later evolved into the modern Internet.

TCP/IP Model is the practical model used on the Internet.

OSI Model is a reference model.

TCP/IP LAYER ARCHITECTURE

Some books describe the TCP/IP model as 4 layers, while others describe it as 5 layers by separating the Physical and Data Link layers.

Five-Layer TCP/IP Model

5. Application Layer

4. Transport Layer

3. Internet Layer/Network layer 2. Data Link Layer

1. Physical Layer

Four-Layer TCP/IP Model

4. Application Layer

3. Transport Layer

2. Internet Layer

1. Network Access Layer

Exam Point

For most competitive examinations:

TCP/IP Model = 4 Layers

OSI = 7 Layers

1. NETWORK ACCESS LAYER

The Network Access Layer is the lowest layer of the TCP/IP Model.

It combines the functions of:

- Physical Layer (OSI)
- Data Link Layer (OSI)

Main Functions

- Data transmission between devices on the same network.
- Encapsulation of IP packets into frames.
- Physical addressing (MAC Address).
- Error detection.

Protocols Used

- Ethernet
- Token Ring
- FDDI (Fiber Distributed Data Interface)
- X.25
- Frame Relay

2. INTERNET LAYER

The Internet Layer corresponds to the Network Layer of the OSI Model.

It is responsible for delivering packets from source to destination regardless of the path taken.

PROTOCOLS OF INTERNET LAYER

1. IP (Internet Protocol)

IP is the most important protocol of the TCP/IP suite.

It provides logical addressing and routing.

Functions

- Assigns IP addresses
- Routes packets
- Delivers packets across networks

2. ARP (Address Resolution Protocol)

ARP is used to find the MAC address of a device when its IP address is known.

Conversion

IP Address → MAC Address

3. RARP (Reverse Address Resolution Protocol)

RARP is used to find the IP address when the MAC address is known. 4. ICMP (Internet Control Message Protocol)

ICMP is used for reporting network errors and problems.

5. IGMP (Internet Group Management Protocol)

Example

Live video streaming to multiple users.

3. TRANSPORT LAYER

The Transport Layer provides end-to-end communication; reliability depends on the protocol, such as TCP or UDP.

PROTOCOLS OF TRANSPORT LAYER

1. TCP (Transmission Control Protocol)

TCP is a reliable, connection-oriented protocol.

2. UDP (User Datagram Protocol)

UDP is a connectionless and unreliable protocol.

Important Fact

Checksum is a 16-bit field used for error detection.

4. APPLICATION LAYER

The Application Layer is the topmost layer of the TCP/IP Model.

It provides services directly to users and applications. Main Functions

- Web browsing
- File transfer
- Email services
- Remote login
- Network management

APPLICATION LAYER PROTOCOLS

1 HTTP (HyperText Transfer Protocol)

Used for accessing websites and web pages.

Example

Opening a website in a browser.

2 FTP (File Transfer Protocol)

Transfers files between computers.

Example

Uploading files to a web server.

3 SMTP (Simple Mail Transfer Protocol)

Used for sending emails.

4 POP (Post Office Protocol)

Used for receiving emails.

5 DNS (Domain Name System)

DNS converts domain names into IP addresses. Example

www.google.com

142.250.183.78

6 DHCP (Dynamic Host Configuration Protocol)

Automatically assigns IP addresses to devices.

Example

When you connect your phone to Wi-Fi, DHCP assigns an IP address automatically.

7 TELNET (Terminal Network)

Allows remote login to another computer.

8 SNMP (Simple Network Management Protocol)

Used for monitoring and managing network devices.

Example

Managing routers and switches.

OSI MODEL VS TCP/IP MODEL

Feature	OSI Model	TCP/IP Model
Developed By	ISO	ARPANET
Number of Layers	7	4 (or 5)
Nature	Reference Model	Practical Model
Usage	Theoretical	Internet
Application	Network Design	Real Communication

MOST IMPORTANT FACTS FOR EXAMS

1.TCP/IP was developed before the OSI Model.

2.TCP/IP was developed by ARPANET.

3.TCP/IP is the model used on the Internet.

4.Network Access Layer combines Physical and Data Link layers.

5.Internet Layer corresponds to the Network Layer of OSI.

6.IP is the most important protocol in TCP/IP.

7.ARP converts IP Address to MAC Address.

8.RARP converts MAC Address to IP Address.

9.ICMP reports network errors.

10.IGMP manages multicast groups.

11.TCP is reliable and connection-oriented.

12.UDP is fast and connectionless.

13.DNS converts names into IP addresses.

14.DHCP automatically assigns IP addresses.

15.SMTP sends emails.

16.POP receives emails.

17.FTP transfers files.

18.HTTP is used for web browsing.